

Martin Špetlík

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Personal Profile

Applied ML researcher and Ph.D. candidate with experience in scientific machine learning, graph neural networks, and advanced generative modeling for medical imaging. Developed VQ-VAE, VQ-GAN, and diffusion-based architectures to generate complex medical 3D CT scans, and built ML-based surrogates for complex physical simulation processes. Strong background in programming, statistics and numerical methods. Proven ability to identify research problems, survey the literature, design and implement tailored ML or computational solutions. Published researcher with an interdisciplinary mindset and a strong motivation to apply ML to challenging real-world problems.

Education

Ph.D. in Applied Sciences in Engineering

Technical University of Liberec

Liberec, Czech Republic

Sep 2020 – Mar 2026 (Expected)

- Supervisor: Assoc. Prof. Dr. Jan Březina (jan.brezina@tul.cz).
- Doctoral Research: Pioneered the application of Multilevel Monte Carlo (MLMC) methods for stochastic modeling in fractured rock, an approach made feasible by the integration of machine learning surrogates.

GeMSS - Generative Modeling Summer School

Eindhoven University of Technology

Eindhoven, The Netherlands

June 2024

- Completed intensive 1-week training focused on generative models including VAEs, GANs, diffusion models, or transformer-based architectures.

Master's Degree in Information Technology

Technical University of Liberec

Liberec, Czech Republic

July 2020

- Thesis: *Reconstruction of Distribution from Estimated Generalized Moments*.
- Graduated Cum Laude; recipient of the Dean's Award for Outstanding Master's Thesis.

Bachelor's Degree in Information Technology

Technical University of Liberec

Liberec, Czech Republic

June 2018

- Thesis: *Application of the Multilevel Monte Carlo Method in Hydrogeology*.
- Graduated Cum Laude; recipient of the Dean's Award for Outstanding Bachelor's Thesis.

Work Experience

Research Assistant

Technical University of Liberec

Liberec, Czech Republic

Sep 2023 – Present

- Architected the machine learning strategy for a generative AI project focused on 3D CT scans of pelvic bones, including model design (VQ-VAE, VQ-GAN, diffusion models), data preprocessing pipelines, and specialized techniques for low-sample, high-dimensional biomedical data.
- Executed large-scale model training and experimentation on HPC systems, optimizing job scheduling, memory usage, and runtime performance.
- Integrated an Unscented Kalman Filter (UKF), using FilterPy, with a nonlinear Richards-equation model for data assimilation, designing the coupling logic, state mappings, and full workflow containerization (Docker) for reproducibility.
- Lead developer and primary maintainer of the open-source `mlmc` Python package (<https://pypi.org/project/mlmc/>).
- Developed and managed the full curriculum, lecture materials, assignments, and exams for the Computer Architecture course; designed seminar content for the Graph Theory course.

Intern

Belgian Nuclear Research Center (SCK CEN)

Mol, Belgium

Feb 2023 – Jul 2023

- Mentor: Dr. Eric Laloy (eric.laloy@sckcen.be).
- Designed and implemented a deep-learning surrogate model for numerical homogenization, including dataset construction, neural architecture selection, validation procedures, and deployment for HPC-based workflows.

Junior Researcher

Technical University of Liberec

Liberec, Czech Republic

Jan 2020 – Dec 2022

- Introduced Graph Neural Networks (GNNs) into the research group to accelerate hydrogeological simulations.
- Architected and implemented ML-based meta-models for integration within Multilevel Monte Carlo (MLMC) frameworks to improve simulation efficiency for non-fractured subsurface models.

Python developer (part-time)

Technical University of Liberec

Liberec, Czech Republic

Jul 2018 – Dec 2019

- Developed the initial version of the open-source `mlmc` Python package, establishing its core architecture and feature set.

Back-End Developer (part-time)

Hotel.cz

- Developed and maintained back-end systems for web applications using PHP, SQL, and Linux servers.

Liberec, Czech Republic

Jan 2017 - Jun 2018

Key Research Projects

Generative AI for 3D CT Scans

Liberec, Czech Republic

Czech Science Foundation, Grant No. 24-10862S

2024 - Present

- Led the machine learning strategy and architectural design for generative modeling of 3D CT scans of pelvic bones.
- Developed a specialized pipeline combining VQ-GAN and Denoising Diffusion Models to address challenges of low-sample, high-dimensional medical-imaging data.

Multilevel Monte Carlo (MLMC) Acceleration via Deep-Learning Surrogates

Liberec, Czech Republic

SGS-2025-3560, SGS-2023-3379; EURAD-2 (Grant Agreement ID 101166718)

2023 - Present

- Principal Investigator of two student grants focused on ML-based acceleration of stochastic simulations; Research Team Member for European and National research grants.
- Designed and deployed convolutional surrogate models to replace computationally expensive numerical homogenization in 2D and 3D fracture-matrix systems, yielding confirmed 100× speedups.

Data Assimilation in Non-Linear Models

Liberec, Czech Republic

Technology Agency of the Czech Republic, SS06010280

2023 - Present

- Responsible for data assimilation using Unscented Kalman Filters (UKF) in a nonlinear Richards-equation model.
- Designed experimental setups and integration strategies for UKF-based approaches applied to nonlinear simulation models.

Open-Source Software Development, UQ, and GNN Integration

Liberec, Czech Republic

Technology Agency of the Czech Republic (TH03010227, TK02010118); EURAD (Grant Agreement ID

847593)

2018 - Present

- Lead contributor to the open-source **mlmc** Python package, responsible for its architecture, development, and integration with external tools.
- Applied Uncertainty Quantification (UQ) methods to groundwater processes and introduced Graph Neural Networks (GNNs) as surrogates for accelerating UQ analyses.
- Ensured software portability, reproducibility, and maintainability for long-term multi-partner scientific projects.

Skills

Deep Learning	Multilayer Perceptrons (MLPs), Convolutional Neural Networks (CNNs), Graph Neural Networks (GNNs), Generative AI (VAEs, GANs, Diffusion Models), Vector Quantization (VQ-VAE, VQ-GAN)
Scientific Computing	Multilevel Monte Carlo (MLMC), Surrogate Modeling, Uncertainty Quantification (UQ), Kalman Filters (including Unscented KF), High-Dimensional Simulation Modeling, High-Performance Computing (HPC; job scheduling, GPU-based training)
Programming & Tools	Python (PyTorch, TensorFlow, Spektral, NumPy, Pandas, Scikit-learn), MLflow, Git, Linux, Docker, SQL, REX
Research & Development	Reproducible research, Experimental design, Model development and evaluation, Workflow automation, Technical documentation, Independent problem-solving, Literature-driven method development
Soft Skills	Team collaboration, Critical thinking, Technical communication, Time management, Teaching and mentoring

Publications

JOURNAL ARTICLES

Convolutional Surrogate for 3D Discrete Fracture-Matrix Tensor Upscaling

Martin Špetlík, Jan Březina

PREPRINT. 2025, URL: https://martinspetlik.github.io/assets/pdf/convolutional_surrogate_3D_dfm_tensor_upscaling_preprint.pdf

Deep learning surrogate for predicting hydraulic conductivity tensors from stochastic discrete fracture-matrix models

Martin Špetlík, Jan Březina, Eric Laloy

Computational Geosciences. Springer, 2024, URL: <https://link.springer.com/article/10.1007/s10596-024-10324-8>

Groundwater Contaminant Transport Solved by Monte Carlo Methods Accelerated by Deep Learning Meta-Model

Martin Špetlík, Jan Březina

Applied Sciences. MDPI, 2022, URL: <https://www.mdpi.com/2076-3417/12/15/7382>

CONFERENCE PROCEEDINGS

Groundwater flow meta-model for multilevel Monte Carlo methods

Martin Špetlík, Jan Březina

CEUR Workshop Proceedings, 2021, URL: <https://ceur-ws.org/Vol-2962/paper40.pdf>

Languages

Czech	Native
English	Full Professional Proficiency
German	Limited Working Proficiency